$$\boxed{\underline{\text{COVARIANZA:}} \quad Cov(X,Y) = E\left[(X-\mu_X)\cdot(Y-\mu_Y)\right]}$$

Otra forma de escribirla:

$$Cov(X,Y) = E\left[X\cdot Y - \mu_X\cdot Y - X\cdot\mu_Y + \mu_X\cdot\mu_Y\right] =$$

$$= E[X\cdot Y] - E[\mu_X\cdot Y] - E[X\cdot\mu_Y] + E[\mu_X\cdot\mu_Y]$$

$$= E[X\cdot Y] - \mu_X\cdot E[Y] - \mu_Y\cdot E[X] + \mu_X\mu_Y$$

$$= E[X\cdot Y] - \mu_X\mu_Y - \cancel{\mu_Y\cdot\mu_X} + \cancel{\mu_X\cdot\mu_Y} = E[X\cdot Y] - \mu_X\mu_Y$$

$$\boxed{Cov(X,Y) = E[X \cdot Y] - \mu_X \mu_Y}$$

$$Cov(X,X) = E[X^2] - \mu_X^2 = Var(X) = \sigma_X^2$$

$$Cov(Y,Y) = Var(Y) = \sigma_Y^2$$

Volvemos al experimento:

$\mu_X = 0 \cdot 0.5 + 1 \cdot 0.5 = 0.5$; $\mu_Y = 0 \cdot 0.4 + 1 \cdot 0.6 = 0.6$

$E[X^2] = 0^2 \cdot 0.5 + 1^2 \cdot 0.5 = 0.5$; $E[Y^2] = 0^2 \cdot 0.4 + 1^2 \cdot 0.6 = 0.6$

$\sigma_X^2 = 0.5 - 0.25 = 0.25$; $\sigma_Y^2 = 0.6 - 0.36 = 0.24$

$$E[X \cdot Y] = \sum_{\alpha=0}^{1} \sum_{\beta=0}^{1} \alpha \cdot \beta \cdot p_{X,Y}(\alpha, \beta) = \cancel{0 \cdot 0 \cdot p_{X,Y}(0,0)} + \cancel{0 \cdot 1 \cdot p_{X,Y}(0,1)} + \cancel{1 \cdot 0 \cdot p_{X,Y}(1,0)} + 1 \cdot 1 \cdot p_{X,Y}(1,1)$$

$$= 0.2$$

$$\mathrm{cov}(X,Y) = 0.2 - 0.5 \cdot 0.6 = -0.1$$

<u>Otro ejemplo</u>: X una v.a. $\rightarrow \mu_X, \sigma_X^2$

$Y = a \cdot X + b \rightarrow \mu_Y = a\mu_X + b$

$\sigma_Y^2 = a^2 \sigma_X^2 \rightarrow \sigma_Y = |a| \sigma_X$

$$E[X \cdot Y] = E[X \cdot (a \cdot X + b)] = E[a \cdot X^2 + b \cdot X] = a E[X^2] + b \mu_X$$

$$Cov(X,Y) = E[X \cdot Y] - \mu_X \mu_Y = a E[X^2] + b\mu_X - \mu_X (a\mu_X + b) =$$

$$= a E[X^2] + \cancel{b\mu_X} - a\mu_X^2 - \cancel{b\mu_X} = a\left(E[X^2] - \mu_X^2\right)$$

$$Cov(X,Y) = a \cdot \sigma_X^2 = \sigma_X \cdot a\sigma_X = sign(a)\sigma_X \cdot \sigma_Y$$

$$Cov(X,Y) = Sign(a) \cdot \sigma_X \sigma_Y$$

$$\boxed{\text{Coeficiente de correlación: } \rho_{X,Y} = \frac{Cov(X,Y)}{\sigma_X \, \sigma_Y}}$$

① $\rho_{X,Y} \in [-1, +1]$

② X e Y están linealmente relacionadas con probabilidad 1

$$\Updownarrow$$

$$\rho_{X,Y} = \pm 1$$